



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

CSE 306 Operating Systems

00: Course Overview

Autumn 2026

Instructor: Hao Li

Words Before The Journey

- Teaching Language (speaking)
 - ~~Completely in Chinese, or mainly in Chinese~~
 - Mainly in English, yet partially in Chinese for assistance?
 - Completely in English?
- Code In Class 外弛内张、外圆内方
 - Be at ease physically
 - ▶ Feel free to interrupt for posing related questions, initiating communications, etc. --- Just note that you had better put up your hand first (to catch my attention) and then continue
 - ▶ Feel free to go to the WC during class, but just keep your action as quiet as possible
 - ▶ Feel free to eat or drink (if not forbidden by SBI) in class, but also just keep quiet
 - Be serious spiritually
 - ▶ Study conscientiously
 - ▶ Respect course arrangements

Words Before The Journey

- Office Hours
 - Wed 8:00-10:00, Thu 8:00-10:00, Thu 14:00-16:00 @Lixi 201-4
- Communication
 - Face-to-face
 - ▶ During office hours
 - ▶ During pre-, inter-, post-class times
 - By email: haoli@ahu.edu.cn
 - ▶ When writing the email **title**, you had better write it as “**CSE 306 OS 2026:** {Title content}” to better catch my attention
 - ▶ If do not reply to your email (perhaps because I have not noticed your email), feel free to remind me face-to-face at proper times
 - Teams code: w7ngtns
 - ~~WeChat~~

Course Overview

■ Course Description

- CSE 306 introduces students to the structure of modern operating systems. The course examines components found in high-use operating systems such as UNIX/Linux and Windows.
- The focus of examination includes understanding the underlying technology of each component, skills required to design and build modules of an operating system kernel, and key algorithms related to operating systems.
- Students should be proficient in explaining how components of an operating system interact with each other and play what kind of roles. Students should also possess the potential to design and implement a simple operating system.

■ Textbook

- *Operating System Concepts (10th edition)* by Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, John Wiley & Sons, Inc., 2018

Major Course Topics

- Operating-system structures
- Process management
- Memory management
- Storage management
- Operating-system examples

Major Course Objectives

- Understand fundamental concepts underlying modern operating systems, including virtual memory and multiprogramming
- Develop working knowledge of OS components, including file systems, I/O subsystem, and CPU scheduler
- Be able to design and implement simplified versions of major OS modules

Course Schedule

■ Two instructors

- Prof. Hao Li

Week 1-11 Tue & Fri (08:00-09:35, 09:50-11:25), by Prof. Li

- Prof. Praveen Tripathi

Week 12-15 Tue (08:00-09:35) (subject to change), by Prof. Li

Week 16-17 Tue & Fri (08:00-09:35, 09:50-11:25) (subject to change), by Prof. Tripathi

Week	Dates	Topic
Week 1	09/08 Tue	• Course Overview • Introduction
	09/11 Fri	• Introduction
Week 2	09/15 Tue	• Operating-System Structures
	09/18 Fri	• Operating-System Structures
Week 3	09/22 Tue	• Processes
	09/25 Fri	• Processes
Week 4	09/29 Tue	• Processes • Quiz 1
	10/2 Fri	• National Day Holiday
Week 5	10/06 Tue	• Threads & Concurrency
	10/09 Fri	• Threads & Concurrency
Week 6	10/13 Tue	• CPU Scheduling
	10/16 Fri	• CPU Scheduling • Quiz 2
Week 7	10/20 Tue	• CPU Scheduling
	10/23 Fri	• Synchronization Tools
Week 8	10/27 Tue	• Synchronization Tools
	10/30 Fri	• Review (Agile Arrangement)
		Mid Exam
Week 9	11/06 Fri	• Synchronization Examples
	11/06 Fri	• Synchronization Examples

Week 10	11/10 Tue	• Deadlocks
	11/13 Fri	• Deadlocks
		Course Project out
Week 11	11/17 Tue	• Main Memory
	11/20 Fri	• Main Memory
Week 12	11/24 Tue	• Virtual Memory • Quiz 3
Week 13	12/01 Tue	• Virtual Memory
Week 14	12/08 Tue	• Virtual Memory
Week 15	12/15 Tue	• Review (Agile Arrangement)
		Course Project collection
Week 16	TBD	• Mass-Storage Structure
Week 17	TBD	• I/O Systems
		Final Exam

Composition of Grade	Percentage	Dates	
Quizzes	20	• 09/29 Tue • 10/16 Fri	• 11/24 Tue
Course Project	20	• Week 10-15	
Mid Exam	30	• Week 8	
Final Exam	30	• Upon contingent information	

Course Evaluation

- Quizzes
 - Periodically we will have brief, in-class quizzes that cover course materials
- Examinations
- Course Project
 - Subject to change

Composition of Grade	Percentage
Quizzes	20
Course Project	20
Mid Exam	30
Final Exam	30